DO NOW · AO1 REVISION

Moments, levers and gears

Forces
Paper 2 · Separate Physics only
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State what is required to cause an object to rotate.**	A force that does not act through the pivot or centre of mass.	☐
2	**Name the turning effect of a force.**	A moment	☐
3 ★	**State which distance is used when calculating the moment of a force.**	The perpendicular distance from the pivot to the line of action of the force	☐
4	**The clockwise and anticlockwise moments about a pivot are equal. State what this tells you about the object.**	It is balanced.	☐
5 ★	**Explain how a lever can reduce the force needed to turn something.**	Pushing further from the pivot gives the same moment with a smaller force.	☐
6	**State the function of a lever.**	To transmit the rotational effect of a force.	☐
7	**State the function of a simple gear system.**	To transmit the rotational effect of a force.	☐
8	**State the equation linking moment, force and distance.**	moment = force × perpendicular distance from the pivot (M = F d)	☐
9	**State the standard unit of the moment of a force.**	Newton-metre (N m)	☐
10 ★	**State the principle of moments for a balanced object.**	Total clockwise moment about the pivot = total anticlockwise moment.	☐
11	**A small gear drives a larger gear. State the effect on the speed of rotation and on the moment.**	The larger gear turns more slowly but with a bigger moment.	☐

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